FTX4056S Test 1 Marking guideline

General guidelines

- ✓ *If you provide two answers for a question where there is only one correct answer, we mark the first one and ignore the second one.*
- ✓ *No marks are awarded for answers that are not backed by workings*

QUESTION 1

If I buy this bond today, 25th September, the settlement date will be t+3 working days, 30th September 2024. This means the bond will trade at the coupon date. Hence the next coupon date will be 31 March 2025 [1]

- *Comment*
  - *30 September 2024 was accepted (because of the way the question was worded)*

The amount of the next coupon will be: Coupon x Index ratio

$$Coupon = \frac{1.875\% \times R10{,}000}{2} = R93.75[\mathbf{1}]$$

CPI index ratio

$$\boldsymbol{Index\ ratio} = \frac{\boldsymbol{REFCPI_{sett\_Date}}}{\boldsymbol{REFCPI_{issue\_Date}}}$$

- ❖ Reference CPI on issue date: is 27th July 2016,
  The reference CPI will be based on CPI 4 months back

$$REFCPI_{\boldsymbol{issue\_Date}} = CPI_j + \left[\frac{(CPI_{j+1} - CPI_j)}{D} \times (t-1)\right]$$

$$CPI_{March\ 2016} + \left[\left[\frac{(CPI_{April\ 2016} - CPI_{March\ 2016})}{31}\right] \times 27 - 1\right]$$

$$77.6 + \left[\frac{(78.30 - 77.60)}{31} \times 26\right] = 78.19\ [3]$$

CPI index ratio

$$\boldsymbol{Index\ ratio} = \frac{\mathbf{115.75}}{\mathbf{78.19}} = \mathbf{1.48}\ [3]$$

The amount of the next coupon will be: 93.75 x 1.48=R138.78 **[2]**

Comment

- ✓ *The CPI on the settlement was given, there was no need to calculate it. However, more than 70% of the students went on calculate the CPI on the settlement date, which was an unnecessary waste of time.*
- ✓ *1 method mark was awarded if the numerator for index ratio was incorrect.*
- ✓ *This question did not require you to value the bond. However, close to 100 students went on to value the bond – again, this was an unnecessary waste of time.*
- ✓ *The poor performance on this question is largely attributed to memorising solutions from past exam papers. Which is why many students went on to value the bond and to calculate the CPI(settlement) when it was given*

QUESTION 2

Advantage: It is not exposed to interest rate risk hence it does not suffer capital losses when interest rate increase [1]

Disadvantage: Its price does not appreciate when interest rates decrease, hence it does not produce capital gains if you want to generate trading profits from a decrease in interest rates [2].

- *Comment:*
  - *On the advantages, it is not sufficient to say, "it does not have exposure to interest rate risk" because that can be viewed as a disadvantage as well.*
  - *The fact that "it does not have exposure to interest rate risk" can be packaged as an advantage or as a disadvantage: It is an advantage because it does not suffer capital losses when interest rates increase and it's a disadvantage because it does not produce capital gains if you want to generate trading profits*

QUESTION 3

Schedule of coupon payments

To be more accurate when calculating periodic coupon payments and discount rates, instead of dividing by 4 you can multiply by ($\frac{Actual}{365}$)

$$Next\ coupon\ payment\ = (Current\ JIBAR + Issue\ spread) \times Face\ value$$

$$18th\ Deccember\ 2024 = \frac{4\%+1.65\%}{4} \times R10{,}000 = R141.25\ \textbf{[1]}$$

The rest of the coupons are based on the forward rates

- ✓ $18th\ March\ 2025 = \left[\frac{\left(1+\frac{6\%}{4}\right)^2}{\left(1+\frac{4\%}{4}\right)^1}\right] - 1] = 2\%\ (periodic) \times 4 = 8\%,$ **[2]**

  $18th\ March\ 2025 = \left[\frac{8\%+1.65\%}{4}\right] \times R10{,}000 = R241.25$**[1]**
- ✓ $18th\ June\ 2025 = \left[\frac{\left(1+\frac{7\%}{4}\right)^3}{\left(1+\frac{6\%}{4}\right)^2}\right] - 1] = 2.25\%\ (periodic) \times 4 = 9\%,$ **[2]**

  $18th\ June\ 2025 = \left[\frac{9\%+1.65\%}{4}\right] \times R10{,}000 = R266.25$**[1]**
- ✓ $18th\ September\ 2025 = \left[\frac{\left(1+\frac{8\%}{4}\right)^4}{\left(1+\frac{7\%}{4}\right)^3}\right] - 1] = 2.75\%\ (periodic) \times 4 = 11\%,$ **[2]**

  $18th\ September\ 2025 = \left[\frac{11\%+1.65\%}{4}\right] \times R10{,}000 = R316.25$**[1]**

Discount rates

- ✓ $Fractional\ perioFractional\ period_{25\ Sept\ to 18\ Dec} = \left[\frac{2.8\%+1.2\%}{4}\right] = 1\%$**[1]**
- ✓ $Second\ period_{(18Dec-18Mar)} = \left[\frac{8\%+1.3\%}{4}\right] = 2.325\%$**[1]**

- ✓ $Third\ period(19March - 18June) = \left[\frac{9\%+1.5\%}{4}\right] = 2.625\%$**[1]**
- ✓ $Fourth\ period(19June - 18Dec) = \left[\frac{11\%+1.6\%}{4}\right] = 3.15\%$**[1]**

Value the bond on the next coupon date (18th December 2024)

$$V_{18Dec24} = 141.25 + \frac{R241.25}{(1.02325)} + \frac{R266.25}{(1.02325)(1.02625)} + \frac{R10{,}316.25}{(1.02325)(1.02625)(1.0315)} = R10{,}154.53\textbf{[3]}$$

Discount to the settlement date

$$Dirty\ Price = \frac{R10{,}154.53}{(1+1\%)^{\frac{84}{91}}} = R10{,}061.69\textbf{[3]}$$

Comment

- ✓ *I was impressed by close to 30 students who valued this floating rate bond using the backward induction method. It was a demonstration of strong understanding to apply what we did under callable bonds to floating rate bonds. That is exactly how we expect you to apply the knowledge.*
- ✓ *The trading spread was not constant and hence you could not apply the annuity formula.*

QUESTION 4

1. A decrease in volatility lowers the required rate of return on a callable bond. Lower volatility reduces the bond's risk because it decreases the likelihood of the bond being called. As the risk of the bond being called decreases, investors demand a lower rate of return [3]

2. Positive convexity for a callable bond
   a. It exhibits positive convexity when interest rates are higher relative to the coupon rate [1][1]. This is the because the call option will be far out of the money and the probability of the bond being called is very low [1], which means the price can move freely without the bond being called which makes it possible to realise higher gains that losses for an equal change in yield.[2]
   b. Illustration [3]
   Dummy bond: yield-to-maturity=5%, Coupon rate=5%, Face value=R1000, Price =R1,000, call price is R1,050. Assume interest are currently at 30% and adjust yields up and down by 2%

Coupon rate	Face value	YTM	Price	Price change
5%	R1,0000	28%	R248,15	R21,03
5%	R1,000	30%	R227,12	
5%	R1,000	32%	R208,79	***(18,33)***

[1] [Comment: it is incomplete to say "when interest rates are higher full stop": "its always relative to the coupon rate"; without relating it to the coupon is not correct if the coupon rate is also high.

Comment:

- ✓ *A complete illustration should show the following (1) three prices; initial price and the price after a decrease and increase in interest rates (ii) a higher price appreciation than the price depreciation (iii) there should be an equal change in yield up and down (iv) the coupon rate which is far from the call price for the price to move freely*
- ✓ *I am not sure why many students got the correct answer for 4(2) but when they got to 4(3) they ended up presenting a demonstration of the fact that "the price of a callable bond increases/decreases less rapidly than a vanilla bond" instead of demonstrating their answer in 4(2)*

3. Valuing a callable bond
   a. Backward induction method: with this method you discount the cashflows per period and per node starting from the maturity date [2]. When you find the values per node you determine whether the bond will be called or not and where necessary adjust the bond value to the call price [1].
   With the derivatives method, you value the bond in two parts; You value it as if it is a vanilla bond and then you value the call option separately using derivative techniques like the Black Scholes model and then you find a difference between these two.[2]

   Comment
   - ✓ *An interest rate tree is not a method of valuing callable bond, its an input when we use the backward induction method*
   - ✓ *The arbitrage free valuation technique is not a separate method. It is part of the backward induction method. With the backward induction method, we apply the arbitrage free valuation technique per node using forward rates*

QUESTION 5

a. These two events will lead to a pure steepening of an upward sloping yield curve [1]. They will also result in reduced risk appetite and higher inflation expectations.
   - The higher inflation expectations arise because the countries involved are major producers of key commodities like oil, gas, and wheat. A war, for example, disrupts the supply of these commodities, driving their prices up, which subsequently affects other prices, pushing inflation higher. As inflation expectations rise, investors demand a higher inflation premium on long-term bonds, which increases yields. Alternatively, investors may avoid long-term bonds, driving their prices down and yields up.
   - A war reduces investors' risk appetite. Conflicts in regions like the Middle East or between Russia and Ukraine, especially when involving the United States and Europe, can create instability in the affected countries and economies that trade with them. This instability disrupts investments and commodity supply chains, affecting multiple sectors that rely on these inputs. Consequently, the uncertainty reduces investors' appetite for risk, prompting a shift from riskier assets, like long-term bonds, to safer ones, such as short-term bonds. Increased demand for short-term bonds raises their prices and lowers their yields, while a sell-off of long-term bonds depresses their prices and drives yields higher.

Comments

- *1 mark for pure steepening*
- *2 marks for explaining the impact on short- and long-term yields*
- *2 marks for stating that these two events increase inflation expectations and reduce risk appetite*
- *2 marks for connecting how the events increase inflation expectations and reduce risk appetite (increased uncertainty and disrupting the supply chain)*

b. These events will lead to higher inflation expectations. When inflation expectations rise, investors become concerned about the impact of inflation on their returns. As a result, they will avoid (or sell) vanilla bonds, which pay a fixed coupon, and instead seek inflation-linked bonds that adjust payments based on actual inflation. Alternatively, investors may demand higher returns on vanilla bonds to compensate for the inflation risk. This shift increases demand for inflation-linked bonds, leading to relatively higher prices and lower yields-to-maturity. Meanwhile, the reduced demand for vanilla bonds, whose fixed cash flows are exposed to inflation risk, drives their prices down and results in relatively higher yields-to-maturity. **[4]**

Comments:

- *You get marks for the section that is "underlined". The first two lines (not underlined) draw from the previous question.*
- *Specify the impact on the value of the inflation-linked bond, the key word is value of the bond.*
- *Explain the demand and price fundamentals for these two bonds. The fact that an inflation-linked bond compensates you for actual inflation does not cause the value to increase. The value increases because of high demand as investors seek refuge in bonds that provide a hedge for inflation*

QUESTION 6

Consumer price index (CPI) numbers or inflation numbers are not issued for a particular day. Statistics South Africa only release the CPI or inflation for a month not a for particular day. This can be addressed by issuing inflation for each day as opposed to monthly. [2]

Inflation figures are not released in real time but they are issued retrospectively. This can be addressed by issuing inflation numbers in real time. [2]

QUESTION 7

1. Four activities in the portfolio management process: setting the investment objectives, developing and implementing a portfolio strategy; monitoring the portfolio (risk management); and adjusting the portfolio.[4]
2. Matching distribution of cashflows allows you to match for large and non-parallel shifts of the yield curve.[3]

   Comment:

   ✓ *This question was specific on why we match this factor. No marks were awarded for general answers like "to ensure that the returns of the portfolio match the returns of the index"*

   ✓ *It's not enough to say this factor "addresses the shortcomings of duration or it is a substitute for convexity" – specify what you capture by matching this factors*

3. Some will avoid pure indexing; its costly and inefficient to implement because you have to buy all the bonds in the index and your net returns will always below the index because there is not room to enhance the returns [2]
   Some would avoid enhanced indexing: You will pay more management fees compared to indexing because there is room for the manager to apply their skill to enhance the returns. Since you buy a sample of bonds your gross returns may differ from the index [2]
   Reasonable answer will be awarded marks
4. You create a bond portfolio whose duration is equal to the duration of the obligation [1]. You then fully fund the obligation by investing an amount that is equal to the present value of the obligation. [2]